Determining Quadrupole Deformation β_2 Using Coulomb Excitation

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Why Nuclear Shape Matters

- Nuclei often exhibit quadrupole deformation.
- Deformation reveals collectivity, shell evolution, and shape coexistence.
- Coulomb excitation provides direct access to charge deformation.

Theory: $B(E2)$, Q_0, and β_2

$$Q_0 = \frac{3}{\sqrt{5\pi}} Z R_0^2 \beta_2$$

$$B(E2) = \frac{5}{16\pi} e^2 Q_0^2$$

$$\beta_2 = \frac{4\pi}{3ZeR_0^2} \sqrt{B(E2)}$$

- Measuring $B(E2)$ gives β_2 directly.

Coulomb Excitation Basics

- Bombard nucleus with electromagnetic field (below Coulomb barrier).
- Pure EM interaction: no nuclear force.
- Observe scattering + γ rays.

Experimental Setup

- Inverse kinematics with RIB.
- High-Z target (Pb/Sn).
- Silicon detectors + HPGe (e.g. MINIBALL).
- Measure γ yields vs. angle.

Methodology

1. Extract γ peaks and correct efficiencies.
2. Normalize using target excitation or elastic scattering.
3. Fit data with GOSIA to obtain E2 matrix elements.
4. Compute $B(E2)$ and convert to β_2.

^{30,32}Mg Results

- $B(E2) = 253(21), 434(52)\ \mathrm{e}^2\mathrm{fm}^4$.
- $\beta_2 = 0.40, 0.50$.

^{74,76}Zn Results

- $B(E2) = 351(24), 285(56)\ \mathrm{e}^2\mathrm{fm}^4$.
- $\beta_2 \sim 0.1$.

^{130}Xe Results

- $\beta \approx 0.17$, $\gamma \approx 23^\circ$.
- Extracted via multi-step Coulex.

Conclusions

- Coulomb excitation gives direct access to $B(E2)$.
- From $B(E2) \rightarrow \beta_2$.
- Technique is powerful for both stable and exotic nuclei.